Convex Optimization 作业 4

October 2025

1 A set C is midpoint convex if whenever two points a, b are in C, the average or midpoint (a + b)/2 is in C. Obviously a convex set is midpoint convex. It can be proved that under mild conditions midpoint convexity implies convexity. As a simple case, prove that if C is closed and midpoint convex, then C is convex.

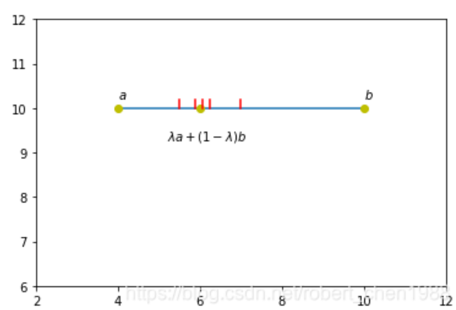

图 1: 图示

任意一点 $\lambda a + (1-\lambda)b$ 可以用 a，b 的二分表示（如图中的红线，不断对区域中切，将中点无限逼近 $\lambda a + (1-\lambda)b$）：

- $\lambda a + (1-\lambda) b \approx \frac{a+b}{2}$
- $\lambda a + (1-\lambda) b \approx \frac{a+\frac{a+b}{2}}{2}$
- $\lambda a + (1-\lambda) b \approx \frac{\frac{a+\frac{a+b}{2}}{2}+\frac{a+b}{2}}{2}$
- $\vdots$
- $\lambda a + (1-\lambda) b \approx \left(1-\frac{m}{2^k}\right) a + \frac{m}{2^k} b$

2 Convex Combination in Convex Sets

Let $C \subseteq \mathbb{R}^n$ be a convex set, with $x_1, \ldots, x_k \in C$, and let $\theta_1, \ldots, \theta_k \in \mathbb{R}$ satisfy $\theta_i \geq 0$, $\theta_1 + \cdots + \theta_k = 1$. Show that $\theta_1 x_1 + \cdots + \theta_k x_k \in C$ for $k \geq 3$.

k=1,2 时显然成立，设 k=n 时条件成立，k=n+1 时

$$\begin{aligned}\theta_1 x_1 + \cdots + \theta_k x_k &= \theta_1 x_1 + \cdots + (1-\theta_1-\theta_2-\cdots-\theta_{k-1}) x_k \qquad (1)\\ &= (\theta_1+\theta_2+\cdots+\theta_{k-1})\left(\frac{\theta_1}{\theta_1+\theta_2+\cdots+\theta_{k-1}} x_1 + \cdots + \frac{\theta_{k-1}}{\theta_1+\theta_2+\cdots+\theta_{k-1}} x_{k-1}\right)\\ &\quad + (1-\theta_1-\theta_2-\cdots-\theta_{k-1}) x_k \qquad (2)\end{aligned}$$

其中

$$\frac{\theta_1}{\theta_1+\theta_2+\cdots+\theta_{k-1}} x_1 + \cdots + \frac{\theta_{k-1}}{\theta_1+\theta_2+\cdots+\theta_{k-1}} x_{k-1} \in C \tag{3}$$

$$\text{because} \frac{\theta_1}{\theta_1+\theta_2+\cdots+\theta_{k-1}} + \cdots + \frac{\theta_{k-1}}{\theta_1+\theta_2+\cdots+\theta_{k-1}} = 1 \tag{4}$$

设

$$y = \frac{\theta_1}{\theta_1+\theta_2+\cdots+\theta_{k-1}} x_1 + \cdots + \frac{\theta_{k-1}}{\theta_1+\theta_2+\cdots+\theta_{k-1}} x_{k-1} \tag{5}$$

$$\theta_1 x_1 + \cdots + \theta_k x_k = (\theta_1+\theta_2+\cdots+\theta_{k-1}) y + (1-\theta_1-\theta_2-\cdots-\theta_{k-1}) x_k \tag{6}$$

因为两项系数相加为 1 且 $y \in C$, $x_k \in C$, 我们有 $\theta_1 x_1 + \cdots + \theta_k x_k \in C$, 得证。

proved

3 Show that the convex hull of a set S is the intersection of all convex sets that contain S

凸包（Convex Hull）：一个集合 S 的凸包是指包含 S 的最小凸集。也就是说，它是包含 S 中所有点的最小凸集。用数学语言来说，如果 S 是平面上的一组点，那么 S 的凸包是一个凸多边形，其顶点是 S 中的一些点，且 S 中的所有点都在这个多边形的内部或边界上。

设 C 为所有包含凸集 S 的交集

设 M 为凸包

若 $x \in M$，则存在 $x_1, x_2 \in S, \theta \in [0, 1]$，使得

$$x = \theta x_1 + (1 - \theta)x_2 \in \text{Conv}(S)$$

易得 $x \in C$，则 $M \subseteq$C

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则 C=M